

MAHARSHI SHUKLA

Data Scientist — AI Engineer — LLM Engineer — Machine Learning Engineer

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PROFESSIONAL SUMMARY

AI-focused Data Scientist and Data Engineer with experience building scalable analytics, machine learning, and automation pipelines using Python, SQL, PySpark, Databricks, Delta Lake, MLflow, and Power BI. Skilled in predictive modeling, anomaly detection, NLP, LLM-based applications, retrieval-augmented generation, LangChain workflows, vector search, data quality validation, and cloud-based analytics solutions. Experienced in transforming complex healthcare, insurance, financial, and business datasets into production-ready insights, dashboards, and machine learning-ready data products.

SKILLS

- **Programming Languages:** Python, SQL, R, Java, Bash/Shell Scripting
- **AI, Machine Learning & Data Science:** Predictive Modeling, Classification, Regression, Anomaly Detection, NLP, Sentiment Analysis, Time Series Forecasting, Feature Engineering, Model Evaluation, Statistical Analysis, Data Mining
- **Generative AI & LLM Engineering:** Large Language Models, Prompt Engineering, RAG, LangChain, Vector Databases, Embeddings, Semantic Search, Conversational AI, AI Agents, Document Q&A, LLM Evaluation
- **Python & ML Libraries:** Pandas, NumPy, Scikit-learn, XGBoost, Matplotlib, Seaborn, Beautiful Soup, Requests
- **Data Science & MLOps Platforms:** Databricks, MLflow, Azure Machine Learning, Jupyter Notebook, Dataiku, Amazon SageMaker
- **Cloud & Big Data:** Azure, AWS, GCP, Snowflake, Apache Spark, PySpark, Delta Lake, Unity Catalog, Kafka
- **Databases & Data Engineering:** MySQL, PostgreSQL, MS SQL Server, MongoDB, SQLite, T-SQL, PL/SQL, Azure Data Factory, Apache Airflow, ETL/ELT, Data Quality Validation
- **Visualization & BI:** Microsoft Power BI, Tableau, DAX, Matplotlib, Seaborn, Dashboard Automation

WORK EXPERIENCE

H&B Analyst I

June 2025 – Current

Aon, Toronto

- Reduced manual reporting effort by 50% by automating ingestion, transformation, and validation of 5,000+ insurance experience reports from Canada Life, Desjardins, and Sun Life using **Databricks, PySpark, SQL, Python**, and **Delta Lake**.
- Improved data reliability by 35% by designing a **Unity Catalog**-enabled lakehouse architecture with governed Delta tables, schema validation, metadata tracking, audit logging, and reusable quality control workflows.
- Increased data integrity by 40% by implementing automated validation checks for duplicate records, missing values, invalid dates, schema drift, outliers, and inconsistent carrier reporting formats using **Python, SQL**, and **PySpark**.
- Enabled machine learning-ready analytics by engineering structured datasets from claims, premiums, benefits, loss ratios, client attributes, carrier files, and reporting periods to support anomaly detection, trend analysis, and predictive modeling.
- Built anomaly detection workflows to identify unusual claim patterns, irregular cost trends, and high-risk reporting anomalies by applying **scikit-learn**, statistical analysis, feature engineering, and **MLflow** experiment tracking in Databricks.
- Accelerated executive decision-making by developing automated **Power BI** dashboards for claims analytics, premium trends, loss ratio monitoring, client performance, and business growth using DAX, dynamic slicers, drillthrough pages, and scheduled refresh logic.
- Reduced repetitive analysis work by applying AI and NLP-driven automation concepts to extract structured insights from carrier reports, claims data, dashboard outputs, and business performance metrics.

Data Engineer

May 2021 – September 2022

Shree Drashti Infotech, India

- Reduced data latency by 20% by building and optimizing **SQL-based ETL pipelines** for real-time and historical data processing across analytical reporting workflows.
- Increased actionable data availability by 30% by integrating structured and semi-structured data sources into enterprise data pipelines for business intelligence, customer behavior analysis, and predictive analytics.
- Improved reporting accuracy to 99% by developing automated validation scripts, reconciliation checks, and data quality controls using **SQL** and **Python**.
- Enabled faster KPI monitoring by designing analytical datasets and **Power BI** dashboards for customer trends, campaign performance, operational metrics, and business reporting.
- Improved pipeline scalability and query performance by optimizing SQL logic, indexing strategies, transformation workflows, and ingestion processes for production analytics use cases.
- Partnered with analytics and data science teams to prepare machine learning-ready datasets, identify high-value data sources, and support downstream modeling and reporting requirements.

AI & DATA SCIENCE PROJECTS

LLM-Powered Document Intelligence and RAG Assistant

Project

- Reduced manual document review effort by designing a **LangChain**-based RAG assistant using LLMs, embeddings, semantic search, vector databases, prompt engineering, and context-aware question answering.
- Improved information retrieval quality by building a modular pipeline for document loading, text chunking, embedding generation, vector indexing, similarity search, prompt construction, and LLM-based response generation.
- Reduced hallucination risk by engineering prompts that grounded LLM responses in retrieved document context and generated structured answers for business reporting, knowledge retrieval, and document analysis workflows.
- Enabled conversational AI use cases by designing reusable components for document summarization, entity extraction, question answering, and insight generation across unstructured files.

Machine Learning-Driven Cryptocurrency Trading System

September 2024 – December 2024

- Achieved 82% and 80% prediction accuracy by developing **XGBoost** and **Random Forest** models to forecast cryptocurrency market trends using historical price, volume, and market movement indicators.
- Improved trading signal quality by engineering technical indicators, market-based features, trend signals, and sentiment variables for model training and evaluation.
- Enhanced model-driven decision-making by integrating **NLP-based sentiment analysis** using the **VADER** algorithm to analyze market sentiment from textual data.
- Accelerated performance monitoring by designing an interactive dashboard to track market movements, trading performance, prediction outputs, and sentiment-based insights in real time.

Health Risk Analysis Using Smoking and Drinking Dataset

September 2023 – December 2023

- Analyzed 991,000+ healthcare records to identify relationships between smoking, drinking, BMI, blood pressure, age, and health risk indicators using **Python**, **Pandas**, **NumPy**, and statistical analysis.
- Improved data usability by cleaning, transforming, and preparing large-scale structured healthcare data using Python-based preprocessing and Azure cloud services.
- Generated actionable healthcare insights by applying data mining techniques to uncover health patterns, risk factors, correlations, and demographic-level trends.
- Improved business and healthcare interpretation by building interactive **Power BI** dashboards to visualize health risk trends, statistical patterns, and population-level insights.

Bone Fracture Detection Using Machine Learning and Deep Learning

August 2021

- Improved medical image classification workflows by developing and evaluating multiple **machine learning** and **deep learning** models for bone fracture detection using healthcare image data.
- Identified the best-performing fracture classification approach by comparing traditional ML models and deep neural network architectures using model evaluation metrics.
- Applied AI-based healthcare analytics concepts to support automated medical image interpretation and diagnostic decision-making.
- Published research focused on using artificial intelligence and deep learning techniques to improve healthcare diagnostics and bone fracture detection.

PUBLICATIONS

- **Optimizing In-Database Analytics for Dynamic Data Exploration and Predictive Insights**
- **Bone Fracture Detection Using Deep Learning**

EDUCATION

MS in Data Analytics Engineering

August 2023 – May 2025

Northeastern University, Vancouver, BC

BE in Computer Engineering

August 2017 – May 2021

Gujarat Technological University, India

CERTIFICATIONS

- **Microsoft Azure Data Engineering Professional**
- **IBM Data Science Professional**
- **Databricks Fundamentals**